Econ 602 Notes

Parker Howell

Last Updated: 2024-02-08

Simultaneous-move Games of Complete Information

Can model these as strategic-form games

Def "strategic-form game"

A strategic-form game is a $G = (S_i, u_i = 1, ..., I)$ such that

1. $\{1, ..., I\}$ is a set of players
2. S_i is a set of strategies for player i
3. $u_i : \Pi_{j=1}^{I} S_j \to \mathbb{R}$ is a payoff function for player i. (u_i maps from strategy profiles to real (integrable) numbers.)

Example 1: introduce NE

Def "Nash Equilibrium (PSNE)"

Call $(s_1^*, ..., s_I^*)$ a Nash equilibrium if, for each $i = 1, ..., I$

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}), \forall s_i \in S_i.$$

"deviating to some s_i is not going to make me any better off.

Def "Best Response"

Say $s_i \in S_i$ is a best response for i under s_{-i} if

$$u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i}) \quad \forall s_i' \in S_i.$$

Def "Best Response Correspondence"

$BR_i : S_{-i} \rightrightarrows S_i$ such that

$$BR_i(s_{-i}) = \{s_i \in S_i : s_i \text{ is a best response for } i \text{ under } s_{-i}\}$$

Remark 1 A strategy profile is a Nash Equilibrium if and only if, for each $i = 1, ..., I$

$$s_i^* \in BR_i(s_{-i}).$$

Example 2 Maximize revenue to find BR correspondence. Determine PSNE from correspondence.

Example 3 "Increasing payoffs can hurt"

Difference between Game Theory and Decision Theory: In DT, increasing payoffs cannot hurt. In GT, there are scenarios under which increasing payoffs can hurt.

Example 4 "incentive to hide what you are playing"

Randomization is a way of hiding what's being played.

Def "set of mixed strategies of i"

The set of probability measures or distributions on S_i is $\Delta(S_i)$.

Given $E_i \subseteq S_i$ (event), $\sigma(E_i)$ is the probability that i plays a strategy in E_i, if i chooses the mixed strategy σ_i.

Remark 2 "infinite strategies have no atoms..." If S_i is finite, then $\sigma_i(E_i) = \sum_{s_i \in E_i} \sigma_i(s_i)$. But, this property need not hold when S_i is infinite.

Remark 3 We have (above) defined a new game called the "mixed extension" of $G = (S_i, u_i : i = 1, ..., I)$. The differences are (1) strategy sets are now $\Delta(S_i)$, and (2) the payoff of i from $(\sigma_1, ..., \sigma_I)$ is $u_i(\sigma_1, ..., \sigma_I)$.

Remark 4 Consider a finite game (a game in which S_i is finite for each $i = 1, ..., I$.). The payoff is

$$u_i(\sigma_1, ..., \sigma_I) = \Sigma_{S_I} \cdots \Sigma_{S_1} u_i(s_1, ..., s_I)\sigma_1(s_1) \cdots \sigma_I(s_I).$$

Def "MSNE"

Call $(\sigma_1, ..., \sigma_I)$ a "Mixed Strategy Nash Equilibrium" (MSNE) if, and only if, for each $i = 1, ..., I$

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}), \forall \sigma_i \in \Delta(S_i).$$

Lemma 1 "MSNE $\iff$ weakly better than all pure strategies"

A strategy profile $(\sigma_1, ..., \sigma_I)$ is a MSNE if, and only if, for each i

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(s_i, \sigma_{-i}^*), \forall s_i \in S_i.$$

Lemma 2 "MSNE $\iff$ strategy in support is best"

Fix a finite game $G = (S_i, u_i : i = 1, ..., I)$.

Then $(\sigma_1, ..., \sigma_I)$ is a MSNE if, and only if, for each i and for each $s_i \in S_i$, with $\sigma_i^*(s_i) > 0$,

$$u_i(s_i, \sigma_{-i}^*) \geq u_i(r_i, \sigma_{-i}^*), \forall r_i \in S_i.$$

Remark 5 "Lemma 2 doesn't have the right context for the infinite case"

Cor 1 "indifference condition"

Fix a finite game $G = (S_i, u_i : i = 1, ..., I)$.

If $(\sigma_1, ..., \sigma_I)$ is a MSNE and with $\sigma_i^*(s_i), \sigma_i^*(r_i) > 0$, then

$$u_i(s_i, \sigma_{-i}^*) = u_i(r_i, \sigma_{-i}^*).$$

Example 5 "Lemma 2 doesn't work in the infinite case + intuition for how to extend"

Thm "from online notes"

1. If σ_i induces F_i, then F_i is a CDF
2. If F_i is a CDF, then there exists a *unique* σ_i that induces F_i.

Remark 6 "unanswered questions"

Remark 7 "pdf/cdf technical notes" omitted

Thm "Kakutani's fixed point thm"

Let X be nonempty compact convex set. If F is nonempty, convex-valued, and has a closed graph, then F has a fixed point.

Term "Mixed Best Response Correspondence" $\tilde{BR}_i : \Pi_{j \neq i} \Delta(S_j) \rightrightarrows \Delta(S_i)$ such that

$$\tilde{BR}_i(\sigma_{-i}) = \{\sigma_i \in \Delta(S_i) : \sigma_i \text{ is a best response under } \sigma_{-i}\}.$$

Thm 1 "conditions to guarantee existence of PSNE"

Fix a game where, for each $i = 1, ..., I$, if

1. there exists an n_i such that S_i is a nonempty, compact, convex subset of $\mathbb{R}^{n_i}$,
2. $u_i : S \to \mathbb{R}$ is continuous, and
3. for each s_{-i}, holding s_i fixed, $u_i(\cdot, s_{-i}) : S_i \to \mathbb{R}$ is quasiconcave.

   nonempty: from S_i being compact and continuity of utility fn (BR has to be nonempty)

   closed-graph: from continuity of u_i

   convex-valued (F): from quasiconcavity of $u_i(\cdot, s_{-i}) : S_i \to \mathbb{R}$

Thm 2 "conditions to guarantee existence of MSNE"

For each finite game, there exists a MSNE.

Thm 3 "Glicksberg's thm"

Fix a game, G where, for each i,

1. S_i is compact metric space, and
2. $u_i : S \to \mathbb{R}$ is continuous.

Then G has a MSNE.

Remark 8 Glicksberg's thm doesn't tell us that the assumptions of compactness and continuity are interesting. E.g., first-price auctions are discontinuous.

Remark 9 Glicksberg's thm only tells us there is some equilibrium–there may be many.

Remark 10 Existence results are most useful if there are qualitative results of how equlibria vary with interesting parameters of the model/game.

Rationalizability and Iterated Dominance

Example 1 "motivation for beliefs: can have wrong beliefs now."

Term "belief" Belief of player i: $\mu_i \in \Delta(S_{-i})$

Def "best response with beliefs"

Fix a nonempty set $Q_i \subseteq S_i$. Say s_i *is a best response under* μ_i *given* Q_i if $s_i \in Q_i$ and

$$\int_{S_i} u_i(s_i, s_{-i}) d\mu_i \geq \int_{S_i} u_i(r_i, s_{-i}) d\mu_i, \ \forall r_i \in Q_i.$$

Say s_i *is a best response under* μ_i if it is a best response under μ_i given S_i.

Def "justifiable"

Fix a nonempty (candidate subset) $Q_i \subseteq S_{-i}$. Say s_i *is justifiable given* Q_i if there exists $\mu_i \in \Delta(S_{-i})$ such that

1. s_i is a best response under μ_i and
2. $\mu_i(Q_{-i}) = 1$.

Say s_i is *justifiable* if it is justifiable given S_{-i}.

Def "rationalizable sets"

Let $R_i^0 = S_i; R_{-i}^0 = S_{-i}$.

Inductively define sets R_i^m and R_{-i}^m such that

- $R_i^{m+1} = \{s_i \in S_i : s_i \text{ is justifiable given } R_{-i}^m\}$, and
- $R_{-i}^{m+1} = \Pi_{j \neq i} R_j^{m+1}$.

Additional notation:

- $R_i^\infty = \cap_{m \geq 0}$ is the set of rationalizable strategies for i.
- $R^\infty = \Pi_{i=1}^I R_i^\infty$ is the set of rationalizable strategy profiles.
- $R^m = \Pi_{i=1}^I R_i^m$ is the set of $m-$rationalizable strategy profiles.

Example 2 "which strategies are justifiable?"

Example 3 "which strategies are justifiable?"

Example 4 "which strategies are justifiable?"

Def "(un)dominated"

Fix $Q_1, ..., Q_I$ where each $Q_j \subseteq S_j$.

Say s_i *is dominated given* $\Pi_{j=1}^I Q_j$ if $s_i \in Q_i$ and there exists a $\sigma_i \in \Delta(S_i)$ such that

$$u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i}),\ \forall s_{-i} \in \Pi_{j \neq i}^I Q_j.$$

Say s_i *is undominated given* $\Pi_{j=1}^I Q_j$ if it is not dominated given $\Pi_{j=1}^I Q_j$.

Say s_i *is (un)dominated* if it is (un)dominated given S.

Def "undominated sets"

Let $U_i^0 = S_i$.

Inductively define U_i^m such that

$$U_i^{m+1} = \{s_i \in S_i : s_i \text{ is justifiable given } R_{-i}^m\}.$$

Additional notation:

- $U_i^\infty = \cap_{m \geq 0} U_i^m$ is the set of undominated strategies for i.
- $U^m = \Pi_{i=1}^I R_i^m$ is the set of $m-$undominated strategy profiles.
- $U^\infty = \Pi_{i=1}^I U_i^\infty$ is the set of *iteratively undominated* (IU) strategy profiles.

Thm 1 "$R^m = U^m$"

Fix a finite game. $R^m = U^m,\ \forall m$.

Prop 1 "justifiable $\iff$ undominated"

Fix a finite game. A strategy s_i is justifiable if, and only if, it is undominated.

Def "Hyperplane"

A hyperplane in $\mathbb{R}^n$ is a set of the form

$$H[\vec{p}, \alpha] = \{\vec{x} \in \mathbb{R}^n : \vec{p} \cdot \vec{x} = \alpha\}$$

for $\vec{p} \in \mathbb{R}^n \setminus \{0\}$ and $\alpha \in \mathbb{R}$.

Def "supporting hyperplane"

Fix $A \subseteq \mathbb{R}^n$ and $\vec{x} \in A$. Say $\vec{p} \in \mathbb{R}^n$ (with $\vec{p} \neq \vec{0}$) *supports A at $\vec{x}$* if

$$\vec{p} \cdot \vec{x} \geq \vec{p} \cdot \vec{y}, \forall y \in A.$$

Say $H[\vec{p}, \alpha]$ supports A at $\vec{x}$ if $\vec{p}$ supports A at $\vec{x}$ and $\vec{p} \cdot \vec{x} = \alpha$.

Thm "Supporting Hyperplane Theorem"

Let $A \subseteq \mathbb{R}^n$, convex and $\vec{x} \in A$. There exists some $\vec{p} \in \mathbb{R}^n$ (with $\vec{p} \neq \vec{0}$) such that $\vec{p}$ supports A at $\vec{x}$ if and only if $\vec{x} \notin int(A)$. (i.e., $\vec{x} \in$ boundary(A))